

Cloud Computing Experiments

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analuse all this 5 experiment give aim procedure , commands if required and result in compact and exam friendly form.

Experiment 1 – Private Cloud using OpenStack (MicroStack)

Aim

To install and configure a Private Cloud using OpenStack (MicroStack) and deploy virtual machine instances with compute, networking, and storage services.

Experient 1 Private Cloud using...

Procedure

1. Install MicroStack on Ubuntu 22.04.
2. Initialize OpenStack services.
3. Open Horizon Dashboard in browser.
4. Login using admin credentials.
5. Create and manage VM instances.

Commands

Bash

```
sudo snap install microstack --beta
```

```
snap list microstack
```

```
sudo apt install python-is-python3
```

```
sudo microstack init --auto --control
```

```
# Get dashboard password
```

```
sudo snap get microstack config.credentials.keystone-password
```

Important OpenStack Services

- Nova – Compute
- Neutron – Networking
- Keystone – Identity
- Glance – Image Management
- Horizon – Dashboard

Experiment 1 Private Cloud using...

Result

Successfully created a single-node private cloud using OpenStack (MicroStack) with working compute, networking, and storage services.

Experiment 1 Private Cloud using...

Experiment 2 – Public Cloud PaaS using Azure App Service

Aim

To develop and deploy a simple web application using Microsoft Azure App Service to understand the concept of Platform as a Service (PaaS).

Experiment 2 Public Cloud PaaS

Procedure

1. Create Azure Student account.
2. Create a simple HTML application.
3. Create Azure App Service.
4. Deploy application using ZIP deployment.
5. Run the application in browser.

Sample HTML Code

```
HTML
<!DOCTYPE html>
<html>
<head>
<title>My Azure PaaS App</title>
</head>
<body>
<h1>Hello from Microsoft Azure PaaS!</h1>
<p>This app is deployed using Azure App Service.</p>
</body>
</html>
```

Steps

- Create Resource Group: **RG-PaaS-Lab**

- Create App Service
- Select Free Tier (F1)
- Upload ZIP file through Deployment Center
- Access application URL

Result

Successfully deployed a simple web application using Azure App Service, demonstrating the concept of PaaS.

Experiment 2 Pubic Cloud PaaS

Experiment 3 – Hybrid Cloud using Azure VPN

Aim

To configure a Hybrid Cloud environment by connecting an on-premises network with Azure cloud using Site-to-Site VPN.

Experiment No 3 Hybrid Cloud

Procedure

1. Create Resource Group.
2. Create Virtual Network (VNet).
3. Create App Subnet and Gateway Subnet.
4. Create Virtual Network Gateway.
5. Create Local Network Gateway.
6. Configure Site-to-Site VPN connection.
7. Verify VPN status.

Network Configuration

Component	Configuration
VNet	10.1.0.0/16
App Subnet	10.1.1.0/24
Gateway Subnet	10.1.255.0/27
On-Prem Network	192.168.1.0/24

Important Settings

- Gateway Type: VPN
- VPN Type: Route-Based
- Connection Type: Site-to-Site (IPsec)
- Shared Key: **Azure123**

Result

Successfully configured a hybrid cloud environment with Azure VPN Gateway and simulated on-premises connectivity.

Experiment No 3 Hybrid Cloud

Experiment 4 – Backup Automation using Azure Blob Storage

Aim

To implement automated backup using Azure Blob Storage and scheduling tools.

Exp NO 4 Azure Blob Storage

Procedure

1. Create Azure Storage Account.
2. Create Blob Container.
3. Login using Azure CLI.
4. Upload backup files.
5. Create PowerShell backup script.
6. Schedule automation using Windows Task Scheduler.

Commands

Azure Login

Bash

```
az login
```

```
az account show
```

Upload Files

Bash

```
az storage blob upload-batch \  
  --account-name mystoragebackup123 \  
  --destination backups \  
  --source C:\backup-data
```

PowerShell Backup Script

PowerShell

```
$source = "C:\backup-data"
$container = "backups"

az storage blob upload-batch `
  --account-name mystoragebackup123 `
  --destination $container `
  --source $source
```

Timestamp Backup

PowerShell

```
$date = Get-Date -Format "yyyy-MM-dd"
$destination = "backups/$date"

az storage blob upload-batch `
  --account-name mystoragebackup123 `
  --destination $destination `
  --source C:\backup-data
```

Scheduling

Use Windows Task Scheduler:

```
powershell.exe
-ExecutionPolicy Bypass -File "C:\scripts\backup.ps1"
```

Result

Successfully implemented automated backup using Azure Blob Storage with scheduling support.

Exp NO 4 Azure Blob Storage

Experiment 5 – VPS / Virtual Machine Setup and Linux Configuration

Aim

To create a VPS/Virtual Machine, install Linux OS, and configure networking settings such as IP address, gateway, and DNS.

Exp NO 5 VPS

Procedure

1. Install VirtualBox or VMware.
2. Create Virtual Machine.
3. Attach Ubuntu ISO.
4. Install Linux OS.
5. Configure DHCP networking.

6. Configure Static IP using Netplan.
7. Verify network connectivity.

VM Configuration

Component	Value
RAM	2 GB
CPU	1–2 Cores
Disk Size	20 GB
OS	Ubuntu 64-bit

Commands

Check IP Address

Bash
`ip a`

Check Connectivity

Bash
`ping google.com`

Edit Netplan Configuration

Bash
`sudo nano /etc/netplan/01-netcfg.yaml`

Sample Static IP Configuration

YAML

```

network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses: [192.168.1.100/24]
      gateway4: 192.168.1.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

Apply Network Changes

Bash
`sudo netplan apply`

Verify Gateway

Bash
`ip route`

Result

Successfully created a virtual machine, installed Linux OS, and configured networking including IP address, gateway, and DNS.

Exp NO 5 VPS